Madhu K M

7204287041 | madhukm826@gmail.com | linkedin.com/in/madhukm | github.com/madhukm

Summary

Embedded Firmware and Verification Engineer with 1.5+ years of experience in firmware validation, NAND Flash–based storage systems, FPGA verification, and embedded software testing at ISRO. Experienced in test case development, white-box testing, failure analysis, VHDL testbench development, and data integrity validation for high-reliability storage systems. Skilled in C, Embedded C, Python scripting, protocol verification, and structured validation workflows aligned with SDLC practices.

Technical Skills

Programming Languages: C ,Embedded C, VHDL,Python.

FPGA & EDA Tools : Vivado Design Suite, Vitis IDE, Libero SoC, ModelSim, STA, bitstream generation, .do scripting

Embedded Platforms : MicroBlaze (soft processor), ARM Cortex-M, PIC16F877A, ESP32, LEON Processor, Kintex UltraScale (KCU105)

Firmware & Validation : Bare-metal, RTOS concepts, interrupt handling, test case development, white-box testing, failure analysis

Storage & Memory : NAND Flash validation, SSR (Solid State Recorder), Flash interface timing, ECC verification

Protocols & Debug Tools: UART, SPI, I²C, RS422 , MIL-STD-1553, USB (EZ-USB FX3)

EXPERIENCE

Graduate Apprentice Trainee – Payload Data Management Group (PDMG) Aug 2025 – Present
URSC, Indian Space Research Organisation (ISRO) *Bengaluru*

• Performed system-level VHDL verification of the FPGA top module — validating interfaces with inter-FPGA communication, NAND Flash, SRAM controller, Flash Controller, and playback modules in a Libero SoC / ModelSim environment, including setup/hold timing margin checks and Static Timing Analysis (STA) across process corners.

• Developed and modified VHDL testbenches for record operation verification under MIN, TYPICAL, and MAX delay conditions using inter-FPGA timing values measured from hardware via oscilloscope; performed waveform analysis and protocol verification for data transfer and timing behaviour.

• Validated a NAND Flash–based Solid State Recorder (SSR) across QM, EM, and FM hardware models — verifying record/playback, file management, and data integrity using MIL-STD-1553 protocol analyzers in Configuration, Operation, and Maintenance modes; performed failure analysis and root-cause investigation for observed anomalies

• Worked with Xilinx KCU105 (Kintex UltraScale FPGA) board for hardware verification and embedded system bring-up; hands-on experience with Vivado design flow including synthesis, implementation, and

• Developed and validated a MicroBlaze soft-processor based system in Vivado — integrating UART, BRAM, and AXI peripherals; used Vitis IDE to write and deploy C firmware for serial communication and peripheral control on FPGA hardware

Firmware Development Intern Jan 2025 – Jul 2025
***B**DGtronix* *Bengaluru*

• *Developed and optimized PIC microcontroller firmware using interrupt-driven design for real-time response.*

• Worked on peripheral-level firmware involving GPIO and timer-based control logic.

• Created structured test plans, validation workflows, and debugging procedures for embedded boards and Performed board bring-up support and troubleshooting using oscilloscopes, logic analyzers, and firmware debug tools.

Projects

Remote Control Panel using PIC16F877A + RS422 + LCD

• Developed embedded firmware for PIC16F877A-based control panel with GPIO interfacing and LCD output.

• Implemented RS422 communication stack with CRC-8 validation for reliable real-time data transfer.

•Enabled serial communication (RS422) between Remote Panel and RHEL 8.1 Workstation using a custom protocol with checksum validation.

• Worked on schematic-level hardware integration, firmware development, and debugging using MPLAB X IDE and simulation tools.

Alcohol Detection and Engine Locking System using Arduino UNO + MQ3 Sensor + GSM + GPS

• Designed real-time firmware to disable ignition within 0.5 seconds upon alcohol detection (MQ3 sensor threshold: 0.4mg/L).

• Focused on sensor interfacing, embedded logic design, and hardware troubleshooting.

- Integrated GSM/GPS modules to transmit alerts (SMS + coordinates) to emergency contacts with 95% transmission success rate .
- Aimed to reduce accidents, promote responsible behavior, and improve road safety through automation.

Education

Dr Ambedkar Institute of Technology Bengaluru, Karnataka
B.E in Electronics and Communication Engineering *Aug 2020 – Aug 2024*
CGPA: 7.46